

DPP-7

- Q.1 The altitude of a parallelopiped whose three coterminous edges are the vectors, $\vec{A} = \hat{i} + \hat{j} + \hat{k}$; $\vec{B} = 2\hat{i} + 4\hat{j} - \hat{k}$ & $\vec{C} = \hat{i} + \hat{j} + 3\hat{k}$ with $\vec{A}$ and $\vec{B}$ as the sides of the base of the parallelopiped, is
(A) $2/\sqrt{19}$ (B) $4/\sqrt{19}$ (C) $2\sqrt{38}/19$ (D) none
- Q.2 Consider ΔABC with $A \equiv (\vec{a})$; $B \equiv (\vec{b})$ & $C \equiv (\vec{c})$. If $\vec{b} \cdot (\vec{a} + \vec{c}) = \vec{b} \cdot \vec{b} + \vec{a} \cdot \vec{c}$; $|\vec{b} - \vec{a}| = 3$; $|\vec{c} - \vec{b}| = 4$ then the angle between the medians $\vec{AM}$ & $\vec{BD}$ is
(A) $\pi - \cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (B) $\pi - \cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$
(C) $\cos^{-1}\left(\frac{1}{5\sqrt{13}}\right)$ (D) $\cos^{-1}\left(\frac{1}{13\sqrt{5}}\right)$
- Q.3 If $A(-4, 0, 3)$; $B(14, 2, -5)$ then which one of the following points lie on the bisector of the angle between $\vec{OA}$ and $\vec{OB}$ ('O' is the origin of reference)
(A) $(2, 1, -1)$ (B) $(2, 11, 5)$ (C) $(10, 2, -2)$ (D) $(1, 1, 2)$
- Q.4 Position vectors of the four angular points of a tetrahedron ABCD are $A(3, -2, 1)$; $B(3, 1, 5)$; $C(4, 0, 3)$ and $D(1, 0, 0)$. Acute angle between the plane faces ADC and ABC is
(A) $\tan^{-1}(5/2)$ (B) $\cos^{-1}(2/5)$ (C) $\operatorname{cosec}^{-1}(5/2)$ (D) $\cot^{-1}(3/2)$
- Q.5 The volume of the tetrahedron formed by the coterminus edges $\vec{a}$, $\vec{b}$, $\vec{c}$ is 3. Then the volume of the parallelepiped formed by the coterminus edges $\vec{a} + \vec{b}$, $\vec{b} + \vec{c}$, $\vec{c} + \vec{a}$ is
(A) 6 (B) 18 (C) 36 (D) 9
- Q.6 Given unit vectors $\vec{m}$, $\vec{n}$ & $\vec{p}$ such that angle between $\vec{m}$ & $\vec{n} = \text{angle between } \vec{p} \text{ and } (\vec{m} \times \vec{n}) = \pi/6$ then $[\vec{n} \vec{p} \vec{m}] =$
(A) $\sqrt{3}/4$ (B) $3/4$ (C) $1/4$ (D) none
- Q.7 $\vec{a}$, $\vec{b}$ and $\vec{c}$ be three vectors having magnitudes 1, 1 and 2 respectively. If $\vec{a} \times (\vec{a} \times \vec{c}) + \vec{b} = 0$, then the acute angle between $\vec{a}$ & $\vec{c}$ is :
(A) $\pi/6$ (B) $\pi/4$ (C) $\pi/3$ (D) $5\pi/12$
- Q.8 If $\vec{a} = \hat{i} + \hat{j} + \hat{k}$, $\vec{b} = 4\hat{i} + 3\hat{j} + 4\hat{k}$ and $\vec{c} = \hat{i} + \alpha\hat{j} + \beta\hat{k}$ are linearly dependent vectors & $|\vec{c}| = \sqrt{3}$, then
(A) $\alpha = 1, \beta = -1$ (B) $\alpha = 1, \beta = \pm 1$ (C) $\alpha = -1, \beta = \pm 1$ (D) $\alpha = \pm 1, \beta = 1$
- Q.9 A vector of magnitude $5\sqrt{5}$ coplanar with vectors $\hat{i} + 2\hat{j}$ & $\hat{j} + 2\hat{k}$ and the perpendicular vector $2\hat{i} + \hat{j} + 2\hat{k}$ is
(A) $\pm 5 (5\hat{i} + 6\hat{j} - 8\hat{k})$ (B) $\pm \sqrt{5} (5\hat{i} + 6\hat{j} - 8\hat{k})$
(C) $\pm 5\sqrt{5} (5\hat{i} + 6\hat{j} - 8\hat{k})$ (D) $\pm (5\hat{i} + 6\hat{j} - 8\hat{k})$

Paragraph for questions nos. 10 to 12

Consider three vectors $\vec{p} = \hat{i} + \hat{j} + \hat{k}$, $\vec{q} = 2\hat{i} + 4\hat{j} - \hat{k}$ and $\vec{r} = \hat{i} + \hat{j} + 3\hat{k}$ and let $\vec{s}$ be a unit vector, then

- Q.10 $\vec{p}$, $\vec{q}$ and $\vec{r}$ are
(A) linearly dependent
(B) can form the sides of a possible triangle
(C) such that the vectors $(\vec{q} - \vec{r})$ is orthogonal to $\vec{p}$
(D) such that each one of these can be expressed as a linear combination of the other two
- Q.11 if $(\vec{p} \times \vec{q}) \times \vec{r} = u\vec{p} + v\vec{q} + w\vec{r}$, then $(u + v + w)$ equals to
(A) 8 (B) 2 (C) -2 (D) 4
- Q.12 the magnitude of the vector $(\vec{p} \cdot \vec{s})(\vec{q} \times \vec{r}) + (\vec{q} \cdot \vec{s})(\vec{r} \times \vec{p}) + (\vec{r} \cdot \vec{s})(\vec{p} \times \vec{q})$ is
(A) 4 (B) 8 (C) 18 (D) 2

One or more than one is/are correct:

- Q.13 Given the following information about the non zero vectors $\vec{A}$, $\vec{B}$ and $\vec{C}$
(i) $(\vec{A} \times \vec{B}) \times \vec{A} = \vec{0}$ (ii) $\vec{B} \cdot \vec{B} = 4$
(iii) $\vec{A} \cdot \vec{B} = -6$ (iv) $\vec{B} \cdot \vec{C} = 6$
Which one of the following holds good?
(A) $\vec{A} \times \vec{B} = \vec{0}$ (B) $\vec{A} \cdot (\vec{B} \times \vec{C}) = 0$ (C) $\vec{A} \cdot \vec{A} = 8$ (D) $\vec{A} \cdot \vec{C} = -9$
- Q.14 Let $\vec{a}$, $\vec{b}$, $\vec{c}$ are non zero vectors and $\vec{V}_1 = \vec{a} \times (\vec{b} \times \vec{c})$ and $\vec{V}_2 = (\vec{a} \times \vec{b}) \times \vec{c}$. If $\vec{V}_1 = \vec{V}_2$ then which of the following hold(s) good?
(A) $\vec{a}$ and $\vec{b}$ are orthogonal (B) $\vec{a}$ and $\vec{c}$ are collinear
(C) $\vec{b}$ and $\vec{c}$ are orthogonal (D) $\vec{b} = \lambda(\vec{a} \times \vec{c})$ when λ is a scalar.
- Q.15 If $\vec{A}$, $\vec{B}$, $\vec{C}$ and $\vec{D}$ are four non zero vectors in the same plane no two of which are collinear then which of the following hold(s) good?
(A) $(\vec{A} \times \vec{B}) \cdot (\vec{C} \times \vec{D}) = 0$ (B) $(\vec{A} \times \vec{C}) \cdot (\vec{B} \times \vec{D}) \neq 0$
(C) $(\vec{A} \times \vec{B}) \times (\vec{C} \times \vec{D}) = \vec{0}$ (D) $(\vec{A} \times \vec{C}) \times (\vec{B} \times \vec{D}) \neq \vec{0}$